Problem A. A Permutation Problem

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

You are given a permutation* p of length n. You may perform the following operation any number of times (including zero):

- Select an integer i such that $2 \le i \le n-1$ and $p_i > \max(p_{i-1}, p_{i+1})$. Then, swap p_{i-1} and p_{i+1}.

A permutation q is reachable from p if and only if there exists a sequence of operations such that performing them on p results in q.

Find the number of permutations reachable from p. Since the number may be very large, output its remainder modulo $998\,244\,353$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 5 \cdot 10^4$). The description of the test cases follows.

The first line of each test case contains an integer n ($3 \le n \le 10^6$), representing the length of p.

The second line contains n distinct integers $p_1, p_2, \ldots, p_n$ ($1 \le p_i \le n$), representing the elements of p.

It is guaranteed that the sum of n over all test cases does not exceed 10^6.

Output

For each test case, output an integer representing the number of permutations reachable from p, modulo $998\,244\,353$.

Example

standard input	standard output
5 3 1 2 3 3 2 1 3 3 2 3 1 8 4 5 3 8 1 6 7 2 5 2 5 4 3 1	1 1 2 12 4

*A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, $[2,3,1,5,4]$ is a permutation, but $[1,2,2]$ is not a permutation (2 appears twice in the array), and $[1,3,4]$ is also not a permutation ($n = 3$ but there is 4 in the array).

Problem B. B1702FC

Input file:	`standard input`
Output file:	`standard output`
Time limit:	4 seconds
Memory limit:	1024 megabytes

An array a is considered ugly if and only if the sum of its elements is a perfect square.

For any array a, define $f(a)$ to be the number of ugly non-empty subarrays† of a.

You are given a positive integer n. Let $M(n)$ be the minimum $f(q)$ over all permutations‡ q of length n.

Construct a permutation p of length n such that $f(p) = M(n)$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^3$). The description of the test cases follows.

The only line of each test case contains an integer n ($1 \le n \le 5000$) — the length of the permutation.

It is guaranteed that the sum of n^2 over all test cases does not exceed 10^8.

Output

For each test case, output n distinct integers $p_1, p_2, \ldots, p_n$ ($1 \le p_i \le n$), representing the permutation p you constructed.

If there are multiple answers, you may output any of them.

Example

standard input	standard output
5 1 2 3 4 5	1 1 2 1 2 3 2 1 4 3 2 5 1 4 3

†An array a is a subarray of an array b if a can be obtained from b by the deletion of several (possibly, zero or all) elements from the beginning and several (possibly, zero or all) elements from the end.

‡A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, $[2, 3, 1, 5, 4]$ is a permutation, but $[1, 2, 2]$ is not a permutation (2 appears twice in the array), and $[1, 3, 4]$ is also not a permutation ($n = 3$ but there is 4 in the array).

Problem C. Cutting Trees

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

You are given an undirected tree consisting of n vertices labeled from 1 to n. The tree is rooted at vertex 1. We define a cutting sequence and its corresponding score of the tree as follows:

- Initialise an empty array a and an integer $s = 0$.
- Perform the following operation until the tree is empty: select a vertex that is still present in the current tree. Then remove all vertices in its subtree[§]. Append the label of the selected vertex to a, and increment s by the number of vertices remaining in the tree after this operation.
- The final a is considered a cutting sequence of the tree, and s is its corresponding score.

Two cutting sequences are different if they have different lengths or if they differ in at least one position.

Find the sum of the scores of all distinct cutting sequences. Since the answer may be very large, output it modulo 998 244 353.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line contains an integer n ($2 \le n \le 3000$) — the number of vertices in the tree.

Each of the next $n - 1$ lines contains two integers u and v ($1 \le u, v \le n$, $u \ne v$), representing an edge between vertices u and v. It is guaranteed that the input forms a valid tree.

It is guaranteed that the sum of n^2 over all test cases does not exceed $9 \cdot 10^6$.

Output

For each test case, output an integer representing the sum of the scores of all distinct cutting sequences, modulo 998 244 353.

Example

standard input	standard output
4 3 1 2 2 3 3 1 2 1 3 4 1 2 2 3 2 4 4 1 2 1 3 3 4	6 10 37 48

[§]A subtree of vertex v is the subgraph of v, all its descendants, and all the edges between them.

Problem D. Divisibility

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

You are given an undirected graph G consisting of n vertices and m edges. You are also given a positive integer k. Note that G may contain self-loops and multiple edges.

For a vertex u, define $f(u)$ to be the smallest non-negative integer d such that d is divisible by k and there exists a path of length d from vertex 1 to vertex u, or $f(u) = -1$ if no such d exists. Note that a path may visit the same vertex or edge more than once.

Find $f(u)$ for every $1 \le u \le n$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 5 \cdot 10^4$). The description of the test cases follows.

The first line of each test case contains three integers n, m, and k ($2 \le n \le 5 \cdot 10^5$, $0 \le m \le 5 \cdot 10^5$, $1 \le k \le 10^9$), representing the number of vertices in G, the number of edges in G, and the constant, respectively.

Each of the next m lines contains two integers u and v ($1 \le u, v \le n$), representing an undirected edge connecting vertices u and v.

It is guaranteed that both the sum of n and the sum of m over all test cases do not exceed $5 \cdot 10^5$.

Output

Output n integers, where the i-th integer represents the value of $f(i)$.

Example

standard input	standard output
7 4 4 3 1 2 2 3 3 4 4 2 5 4 2 1 2 2 3 3 4 4 5 3 3 4 1 2 2 3 3 1 2 1 5 1 2 4 1 2 1 2 2 2 2 1 2 2 2 2 0 1	0 3 3 3 0 -1 2 -1 4 0 4 4 0 5 0 -1 -1 -1 0 2 0 -1

Problem E. Expropriate

Input file:	`standard input`
Output file:	`standard output`
Time limit:	6 seconds
Memory limit:	1024 megabytes

You are given an array a of length n consisting of non-negative integers. You are also given two integers k and m. You must perform the following operation exactly k times:

- Choose an element from a, denote it by x, and remove it. The remaining parts are concatenated. Then, insert $\min(x+1, m)$ at any position in a.

You may choose the same element multiple times in different operations. Find the lexicographically minimum array you can achieve after performing exactly k operations.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 5 \cdot 10^4$). The description of the test cases follows.

The first line of each test case contains three integers n, k, and m ($1 \le n \le 5 \cdot 10^4$, $0 \le k \le n$, $0 \le m \le 2 \cdot n$), representing the length of a, the number of operations to perform, and the parameter for the operation, respectively.

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($0 \le a_i \le m$), representing the elements in a.

It is guaranteed that the sum of n over all test cases does not exceed $5 \cdot 10^4$.

Output

For each test case, output n integers, representing the lexicographically minimum array you can achieve after performing exactly k operations.

Example

standard input	standard output
4 5 2 3 1 2 3 1 2 3 1 6 1 1 1 6 2 2 1 0 0 0 0 0 13 12 14 6 5 4 3 2 0 1 1 7 8 9 10 11	1 1 2 3 3 1 1 2 0 0 0 0 0 2 0 1 1 3 4 5 6 7 7 8 9 10 14

Problem F. Full Alphabet

Input file:	`standard input`
Output file:	`standard output`
Time limit:	10 seconds
Memory limit:	1024 megabytes

In this problem, an alphabet is defined as a permutation of the lowercase English letters from `a` to `z`. A letter x is lexicographically smaller than a letter y under an alphabet A if and only if x appears before y in A.

A string s is lexicographically smaller than a string t under A if and only if either of the following two conditions is satisfied:

- s is a proper prefix of t.
- There exists an integer i ($1 \le i \le \min(|s|, |t|)$) such that $s_j = t_j$ for all $1 \le j < i$, and s_i is lexicographically smaller than t_i under A.

A string s is considered a Lyndon word under an alphabet A if and only if it is strictly lexicographically smaller than all of its non-empty proper suffixes under A.

You are given a string s. Find the number of distinct alphabets A such that s is a Lyndon word under A. Since the answer may be very large, output its remainder modulo 2^{32}.

Input

The only line of each test contains the string s ($2 \le |s| \le 2 \cdot 10^7$).

Output

Output an integer, representing the number of distinct alphabets A such that s is a Lyndon word under A, modulo 2^{32}.

Examples

standard input	standard output
`abcdefghijklmnopqrstuvwxyz`	`2076180480`
`abaabb`	`0`
`sserxhsfan`	`2452619264`
`arcabcagc`	`3598712832`

Problem G. Game on a Graph

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

There is a simple connected undirected graph G consisting of n vertices and m edges, as well as a set of k special vertices $a_1, a_2, \ldots, a_k$. G has an interesting property: all vertices have degree no more than three.

Consider the following game played on G:

- The game involves two players, Alice and Bob. Initially, Alice stands on a vertex s such that s is not special. Then, the players take turns to play, starting with Alice:
  - If it is Alice's turn, suppose she is currently at vertex u. If u is special, the game ends and Alice wins. Otherwise, she chooses a vertex v such that there is an edge between u and v and moves to v. If no such vertex exists, the game ends and Bob wins.
  - If it is Bob's turn, he chooses any remaining edge in G and deletes it. If there are no more edges in G, he does nothing.

  It can be proven that the game will end in a finite number of turns. Throughout the game, the location of Alice is known to both players.

A vertex u is considered winning if and only if:

- u is not special;
- If Alice initially stands on vertex u, she wins the game if both players play optimally.

Find all winning vertices.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 2 \cdot 10^4$). The description of the test cases follows.

The first line of each test case contains three integers n, m, and k ($1 \le k < n \le 2 \cdot 10^5$, $n - 1 \le m \le \min(\frac{3n}{2}, 2 \cdot 10^5)$), representing the number of vertices in G, the number of edges in G, and the number of special vertices, respectively.

Each of the next m lines contains two integers u and v ($1 \le u, v \le n$, $u \ne v$), representing an edge connecting vertices u and v. It is guaranteed that G contains no self-loops or multiple edges. In addition, it is guaranteed that all vertices have degree no more than three.

The next line contains k distinct integers $a_1, a_2, \ldots, a_k$ ($1 \le a_i \le n$), representing the special vertices.

It is guaranteed that both the sum of n and the sum of m over all test cases do not exceed $2 \cdot 10^5$.

Output

For each test case, first output an integer c ($1 \le c \le n - k$), representing the number of winning vertices. Then, output c distinct integers $w_1, w_2, \ldots, w_c$ ($1 \le w_i \le n$), representing the winning vertices. You may output the vertices in any order.

Example

standard input	standard output
4 2 1 1 1 2 1 3 2 1 1 2 2 3 1 4 3 2 1 2 2 3 2 4 3 4 4 5 1 1 2 1 3 2 4 3 4 2 3 4	1 2 1 2 2 1 2 2 2 3

Problem H. Hard Problem

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

You are given an integer n. Find the lexicographically smallest[¶] permutation[‖] p of length n such that $|p_i - p_{(i \bmod n)+1}|$ is not prime for every $1 \le i \le n$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The only line of each test case contains an integer n ($2 \le n \le 2 \cdot 10^5$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, if no such permutation of length n exists, output -1.

Otherwise, output n distinct integers $p_1, p_2, \ldots, p_n$ ($1 \le p_i \le n$), representing the permutation p you found.

Example

standard input	standard output
2 2 3	1 2 -1

[¶]An array a is lexicographically smaller than an array b if and only if one of the following holds:

- a is a prefix of b, but $a \ne b$; or
- in the first position where a and b differ, the array a has a smaller element than the corresponding element in b.

[‖]A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, $[2, 3, 1, 5, 4]$ is a permutation, but $[1, 2, 2]$ is not a permutation (2 appears twice in the array), and $[1, 3, 4]$ is also not a permutation ($n = 3$ but there is 4 in the array).

Problem I. Integer Function

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

For a non-negative integer x, let $f(x)$ be the number of set bits in the binary representation of x.

You are given two integers n and d. Compute:

$$\left(\sum_{i=0}^{n} f(i) \cdot f(i+d)\right) \bmod 998\,244\,353$$

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The only line of each test case contains two integers n and d ($0 \le n, d < 2^{60}$).

Output

For each test case, output one integer representing the value of the expression.

Example

standard input	standard output
5 0 0 1 1 5 7 314 159 114514 1919810	0 1 14 6022 11512513

Problem J. Just Add and Divide

Input file:	`standard input`
Output file:	`standard output`
Time limit:	5 seconds
Memory limit:	1024 megabytes

You are given two arrays a and b of length n consisting of positive integers. You need to answer q queries of three types:

- $1\ i\ x$ ($1 \le i \le n$, $1 \le x \le 10^9$): Assign $a_i \leftarrow x$;
- $2\ i\ x$ ($1 \le i \le n$, $1 \le x \le 10^9$): Assign $b_i \leftarrow x$;
- $3\ l\ r$ ($1 \le l < r \le n$): Compute:

$$\max_{l \le i < j \le r} \frac{a_i + a_j}{b_i + b_j}$$

Queries are processed in the order they are given. Each assignment permanently changes the corresponding array element and affects all subsequent queries.

Input

The first line of each test contains two integers n and q ($2 \le n \le 10^5$, $1 \le q \le 10^5$), representing the length of a and b and the number of queries, respectively.

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$), representing the elements of a.

The third line contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le 10^9$), representing the elements of b.

Each of the next q lines contains a query.

Output

For each query of the third type, it can be proven that the answer can be uniquely represented by an irreducible fraction $\frac{x}{y}$. You should output two integers x and y.

Example

standard input	standard output
6 7 10 20 30 100 1000000000 1000000000 1 5 2 10 1 1000000000 3 1 4 3 2 4 1 3 4 3 1 4 2 1 100 3 1 4 3 5 6	40 3 65 6 10 1 26 3 2000000000 1000000001

Problem K. Konstruct

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

In this problem, a Turing Machine M is defined as a tuple $(Q, q_0, q_h, \delta, p_0)$ where:

- Q: the finite set of all possible states. For simplicity, Q should always be $\{0, 1, 2, \ldots, N-1\}$ for some positive integer N.
- $q_0 \in Q$: the initial state.
- $q_h \in Q$: the halting state.
- $\delta : (Q \setminus \{q_h\}) \times \{0, 1\} \to Q \times \{0, 1\} \times \{-1, 0, 1\}$: the transition function.
- $p_0 \in \mathbb{Z}_{\geq 0}$: the initial position of the pointer.

The Turing Machine operates on a semi-infinite strip of cells c indexed by non-negative integers. Each cell contains either 0 or 1. We denote the value written on cell $i \in \mathbb{Z}_{\geq 0}$ by c_i.

The execution process is governed by the following rules:

- The runtime configuration of M is a pair (q, p), where q is the current state and p is the current pointer position. Initially, $(q, p) = (q_0, p_0)$.
- At each step, the following events occur strictly in order:
  - If $p < 0$, the machine crashes.
  - If $q = q_h$, the machine successfully halts.
  - Otherwise, let $(q_{\text{next}}, w, d) = \delta(q, c_p)$. The machine updates the current cell $c_p \leftarrow w$, transitions to the state $q \leftarrow q_{\text{next}}$, and moves the pointer $p \leftarrow p + d$.

You are given an array a of length n, and an integer k. It is guaranteed that:

- $1 \leq n < 2^{16}$.
- $0 \leq k < 2^{16}$.
- $0 \leq a_i < 2^{16}$ for all $0 \leq i < n$.

All inputs are pre-written on the strip of cells c, encoded in 16-bit little-endian binary. Specifically:

- n is stored in cells 16 to 31.
- k is stored in cells 32 to 47.
- For every $0 \leq i < n$, the value a_i is stored in cells $48 + 16 \cdot i$ to $63 + 16 \cdot i$.
- All other cells on the strip are initially 0.

When x is stored in cells l to r, it means the j-th least significant bit of x is stored at c_{l+j} for $0 \le j \le r-l$. For example, if 10 is stored in cells 1 to 7, then $c_{1..7}$ will be 0101000.

Your task is to construct a Turing Machine M that computes the number of elements in a that are strictly larger than k.

Your machine must eventually reach the state q_h and halt. Upon halting, the answer must be stored in cells 0 to 15. Once the machine halts, the final position of the pointer p and the values written in the rest of the cells do not matter.

Your machine should consist of no more than 11451 states, and halt in no more than $4 \cdot 10^8$ steps.

Input

There is no input for this problem.

Output

First, output an integer N ($1 \le N \le 11451$), representing $|Q|$.

Then, output N lines, each containing 6 integers. The first three integers of the i-th line represent $\delta(i, 0)$, and the following three integers represent $\delta(i, 1)$.

Lastly, output 3 integers q_0, q_h, and p_0 ($0 \le q_0, q_h < N$, $0 \le p_0$).

Example

standard input	standard output
No Input	4 1 1 1 3 1 1 1 1 -1 2 0 1 2 1 -1 0 1 -1 3 0 1 3 1 1 0 3 100

Note

The example input and output are just for demonstration purposes and are incorrect. There will be no input for the actual test cases.

Fun fact: the example output is the Turing Machine for BB(3).

Problem L. Longest Palindromic Substring Queries

Input file:	`standard input`
Output file:	`standard output`
Time limit:	4 seconds
Memory limit:	512 megabytes

You are given a string s of length n. Please answer q queries of the following type:

- You are given two integers l and r ($1 \le l \le r \le n$). Find the length of the longest palindromic substring of the string $s[1..l-1] + s[r+1..n]$, where $+$ represents string concatenation. Here, $s[x..y]$ represents the substring of s from the x-th to the y-th character. Specifically, if $x > y$, then $s[x..y]$ is the empty string.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and q ($1 \le n \le 5 \cdot 10^5$, $1 \le q \le 5 \cdot 10^5$), representing the length of s and the number of queries, respectively.

The second line contains the string s. It is guaranteed that s consists of only lowercase English letters.

Each of the next q lines contains two integers l and r ($1 \le l \le r \le n$), representing a query.

It is guaranteed that both the sum of n and the sum of q over all test cases do not exceed $5 \cdot 10^5$.

Output

For each test case, output q integers, where the i-th integer represents the answer to the i-th query.

Example

standard input	standard output
1 6 21 abcacb 1 1 1 2 1 3 1 4 1 5 1 6 2 2 2 3 2 4 2 5 2 6 3 3 3 4 3 5 3 6 4 4 4 5 4 6 5 5 5 6 6 6	5 3 1 1 1 0 3 2 1 1 1 3 3 2 1 4 3 1 1 1 3